PRINCE CHAUHAN

Senior Quantitative Researcher | Options, Volatility & Systematic Trading

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SUMMARY

Options trader turned quant researcher and developer, with nine years on Indian index derivatives (Nifty, Bank Nifty, Sensex). For the last two years I have built and run a fully automated options desk end to end: market data, research, backtesting, execution, risk and live monitoring, now extended to US index options. Chartered Accountancy training (CA Final, Group 1 cleared) means I care as much about how P&L is measured as how it is made: costs on every leg, no lookahead, overfitting tests on every result, and numbers that tie back to a trade ledger.

EXPERIENCE

SMC Global Securities Ltd. - Senior Quantitative Analyst & Researcher — Dec 2024 - Present

- Built QuantsNerv, the desk's in-house trading platform in Python: live WebSocket tick feeds, real-time IV and Greeks, multi-leg order entry through the DhanHQ API, and browser dashboards for positions, P&L and option-chain flow.
- Run seven systematic strategies in live paper trading since Oct 2026, each with its own daily trade ledger: short straddles, opening-range option selling and buying, Nifty futures breakouts, an IV skew z-score book, a Bank Nifty constituent stat-arb basket and an intraday stock breakout book.
- Combined portfolio backtest, Jan 2021 to Sep 2026 (1,411 sessions of 1-minute data, 0.50% cost on every leg): INR 1.41 Cr net on INR 1 Cr capital, Sharpe 3.67, profit factor 2.09, worst drawdown INR 3.8 L (under 4% of starting capital), 61 of 69 months profitable.
- Built the gate every strategy must pass before going live: walk-forward windows, a parameter-plateau check (neighbouring settings must also make money), double-cost stress, Probabilistic and Deflated Sharpe ratios, Probability of Backtest Overfitting, and a Monte Carlo of drawdowns.
- Wrote the risk layer every order passes through: margin check before entry, daily loss stop, an India VIX gate that stops new option selling when volatility spikes, forced exit at 15:20, slicing to exchange freeze limits, and back-off when the broker rate-limits.
- Built the cost model (TCA) behind every backtest: bid-ask spread that widens with distance from the money, an opening-minutes liquidity penalty, a cap of 5% of 5-minute strike volume per order, and the full Indian charge stack (STT, exchange fees, stamp duty, GST).
- Built a live dealer-gamma (GEX) engine for Nifty, Bank Nifty, Sensex and SPX that marks gamma walls and labels the day as pinned or trending; signals are logged in shadow mode before they touch sizing.
- Extended the desk to US markets: CME futures data through Databento, SPX options through Alpaca, an SPX opening-range bot, an S&P strangles backtest and five years of 5-minute data on ES, NQ, SPX and QQQ.

Smart Bulls Capital - Research Analyst, Derivatives & Quantitative Research — Mar 2017 - Nov 2024

- Traded systematic multi-leg option books on Nifty and Bank Nifty (iron condors, calendars, butterflies), rebalancing delta and vega through the day.
- Built the volatility tools behind those books: option price to IV inversion (Newton-Raphson with a Brent fallback), smile smoothing across strikes, and dealer gamma maps showing where hedging flows flip.
- Built a volatility regime filter (k-means on realised volatility and the IV/RV ratio) that cuts short-gamma size when volatility starts to expand.
- Sized every book off exchange SPAN margin on multi-crore capital and kept drawdowns inside desk limits.

Indian Oil Corporation Ltd.; CA Matta & Associates - Treasury; Audit & Tax — 2014 - 2017

- Treasury cash-flow forecasting at IOCL (2016-2017); statutory audit and corporate tax at CA Matta & Associates (2014-2016).

DESK RESEARCH & BACKTESTING

- Maintain a backtest library of 13 option strategies on 1-minute NSE/BSE data (2019 onward). Each run produces the same four outputs: a PDF tear sheet, a VIX-regime breakdown, an Excel report and a leg-by-leg trade ledger, so any number can be traced to the trades behind it.
- Data checks run before any backtest: tick-to-Parquet reconciliation, coverage audit for missing minutes, a corporate-actions register, and a check that every fill was at a price that actually traded.
- Every strategy carries a record from idea to production: experiment log, a strategy passport, numbers reconciled across tear sheet, Excel and ledger, and an accounting check that P&L ties to the rupee.
- Research tracks that feed the book: India VIX regimes and when short volatility should stand down, IV skew z-scores across strikes, open-interest flow for intraday direction, cash-futures basis, and Johansen cointegration across Bank Nifty constituents.
- Regime and risk models: k-means market regimes, VPIN order-flow toxicity, Monte Carlo VaR, and a genetic parameter search scored on Deflated Sharpe so the search itself cannot overfit.

OPEN-SOURCE RESEARCH CODE

github.com/prince-chauhan-pc/quant-research-portfolio - small Python modules with unit tests and CI on every push. Sample data only; no live strategy logic.

- Point-in-time backtester with margin-based lot sizing, per-leg costs, Sharpe / Sortino / Calmar and a standard trade ledger.
- Volatility surface: Black-Scholes IV inversion, SVI fit per expiry with no-arbitrage checks, Greeks and a dealer gamma (GEX) profile.
- Pairs and baskets (Engle-Granger, half-life, z-score entries), a fill simulator with square-root market impact, and a validation gate with overfitting statistics.

SKILLS

- **Statistics:** Probabilistic and Deflated Sharpe, backtest-overfitting tests, walk-forward and parameter-plateau testing, cointegration (Engle-Granger, Johansen), regime clustering, Monte Carlo VaR.
- **Options:** Pricing and Greeks, IV surfaces (SVI) and skew, dealer gamma (GEX), VIX regimes, SPAN margin sizing.
- **Execution:** Transaction cost analysis, market impact, VPIN, freeze-limit slicing, broker rate limits.
- **Engineering:** Python (pandas, NumPy, SciPy, scikit-learn), asyncio, WebSockets, Flask, SQLite, DuckDB, Parquet, pytest, Git and CI. AI coding agents used inside written test gates.
- **Data & brokers:** NSE / BSE index options, futures and cash; CME futures and US index options; DhanHQ, Databento and Alpaca APIs.

EDUCATION & REGISTRATION

- **B.Com (Honours):** Kirori Mal College, University of Delhi, 2011 - 2014. First Division (71%).
- **CA Final (Group 1 cleared, 78%):** Institute of Chartered Accountants of India, 2018.
- **Registration:** SEBI Registered Research Analyst. NISM Series-XV.